

Supporting Information for *Dynamic drop penetration of horizontally oriented fiber arrays*

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Supplementary Movie Captions

Movie S1: Impact classifications and their associated temporal heat maps.

Movie S2: Drops impact aligned samples of density 144 strands/cm^2 at a velocity $\approx 0.5 \text{ m/s}$. Hydrophobic fibers allow contact line recession and resumption of circular shape whereas hydrophilic fibers enhance the advance of contact line.

Movie S3: Drops impact aligned sample of density 144 strands/cm^2 at a velocity $\approx 0.3 \text{ m/s}$. The first occurrence of the local maximum spread width may not correspond to the absolute maximum spread because spreading occurs non-uniformly in between fiber levels.

Movie S4: Drops impact aligned samples of density 144 strands/cm^2 at a velocity $\approx 0.5 \text{ m/s}$ with time annotations.

Movie S5: Drops at $\sim 5 \text{ m/s}$ impact hydrophilic and hydrophobic fibers.

Supplementary Figures

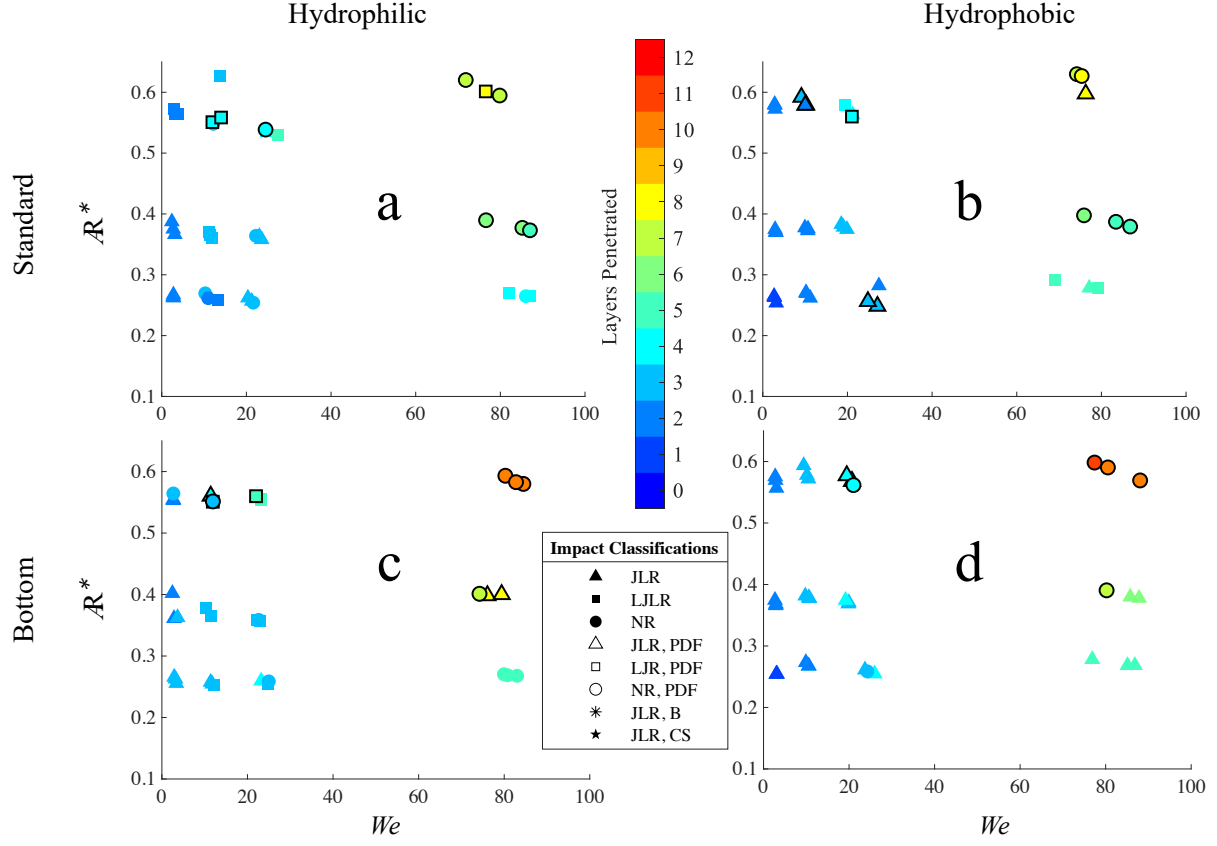


Figure S1: (a-d) Modified aspect ratio versus Weber number plot of staggered fiber arrays. All subplots in the higher row use fibers in Standard orientation whereas all in the lower row use fibers in Bottom orientation. All subplots in the left column use hydrophilic fibers whereas all in the right use hydrophobic fibers. Refer to **Table 2** for classification abbreviations.

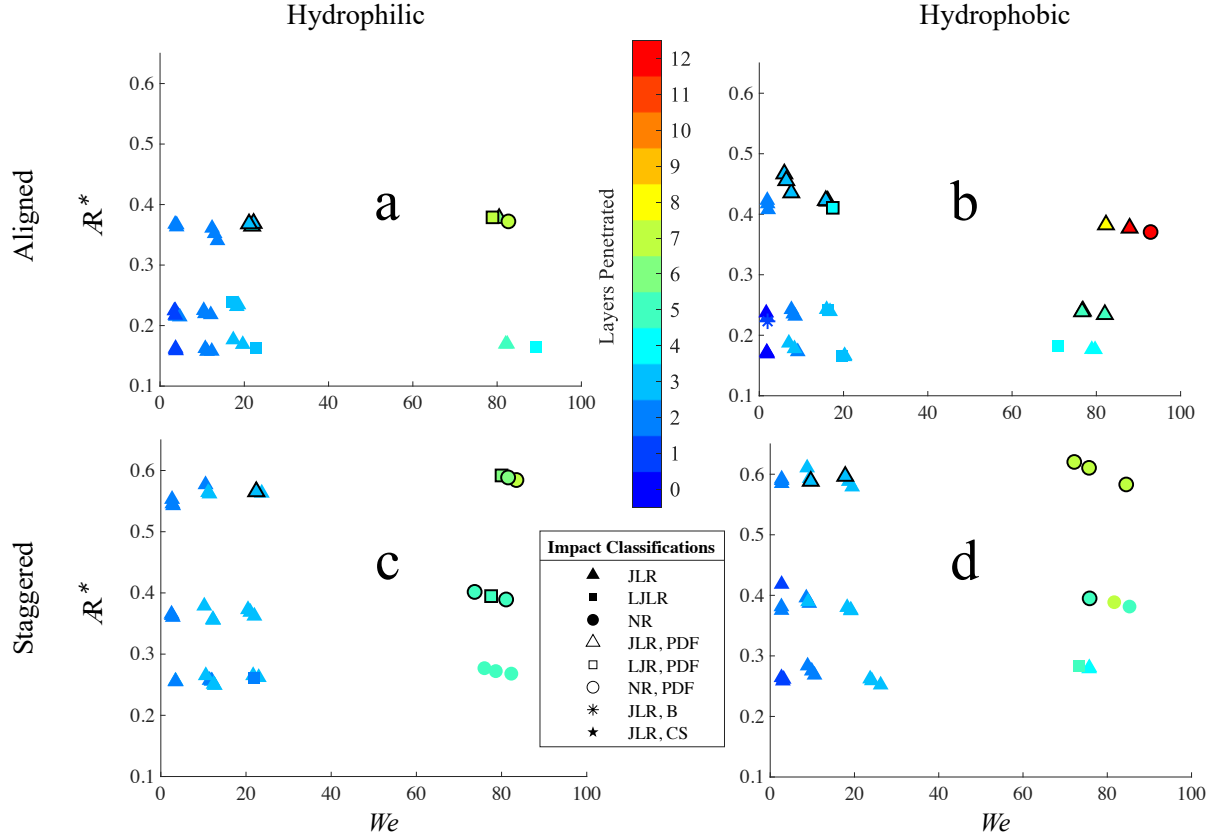


Figure S2: (a-d) Modified aspect ratio versus Weber number plot of fiber arrays in Front/Back orientation. All subplots in the higher row use fibers with aligned configuration whereas all in the lower row use fibers with staggered configuration. All subplots in the left column use hydrophilic fibers whereas all in the right use hydrophobic fibers. Refer to **Table 2** for classification abbreviations.

Auxiliary impact features and volume conservation

Drop impact onto horizontal fiber arrays can be examined well beyond penetration depth. In this section, we explore other impact features, including the lateral spread χ of the drop mass as it deforms within the fiber levels, and the height of the drop exposed above the array d_h .

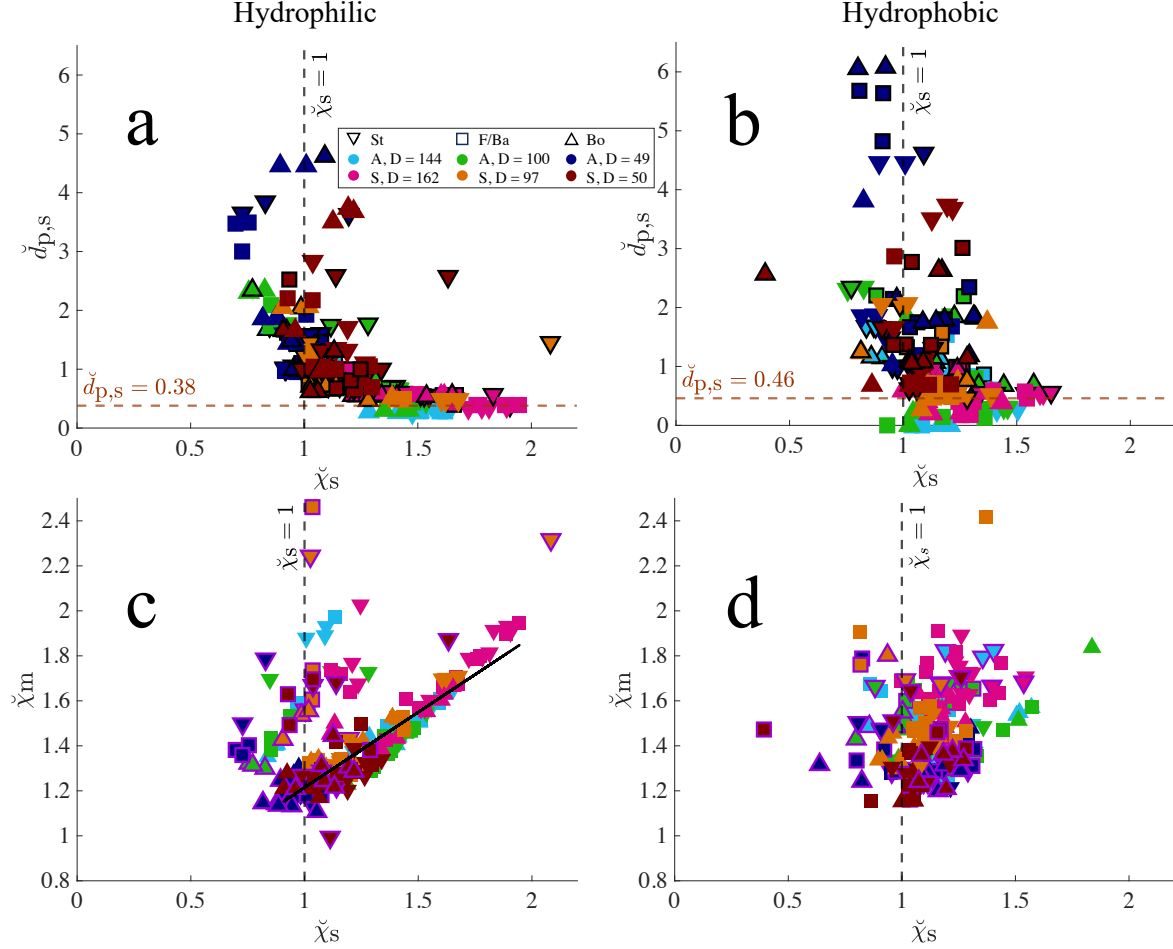


Figure S3: (a,b) Normalized steady-state penetration depth versus normalized steady-state spread. Dashed horizontal lines indicate where $\check{d}_{h,s}$ begins to approach zero (black outline), and drop fragmentation is likely to appear. Fiber density (D) numbers in strands/cm²; St = Standard, F/Ba = Front/Back, Bo = Bottom; A = Aligned, S = Staggered; refer to **Table 2**. (c,d) Normalized maximum spread versus normalized steady-state spread. The marker outline (in black) indicates drop fragmentation. The solid line in panel (c) is drawn for visual reference only. All subplots in the left column use hydrophilic fibers whereas all in the right use hydrophobic fibers.

Lateral spreading restricts penetration depth, not mass infiltration

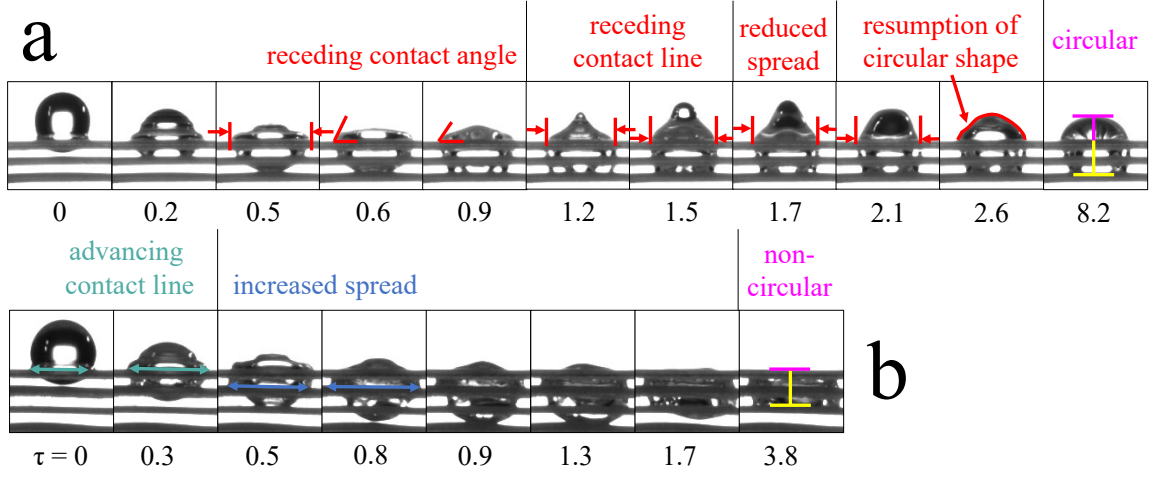


Figure S4: Hydrophobic fibers (a) allow contact line recession whereas hydrophilic fibers (b) enhance the advance of contact line.

When all motion due to impact has ceased, we measure the wetted width of the drop χ_s . Volume conservation dictates that χ_s is negatively correlated with drop penetration of porous media¹. Hydrophobic coatings on our fibers allow the drop, if unbroken, to recede and resume a nearly spherical shape, as demonstrated by **Fig.4b** and **Fig.S4a**. Therefore, the negative correlation between post-impact spreading and penetration is witnessed only for hydrophilic fibers, which promote greater drop spreading via increased advancing contact line mobility, as seen in **Fig.S4b**, **Fig.S3a**, and Movie S2. The receding contact line mobility enabled by hydrophobicity allows data points to cluster around $\check{\chi}_s = 1$ in **Fig.S3b,d**. As previously stated, the reduction in lateral stretching of drops onto hydrophobic surfaces enables greater penetration depths than experienced by equivalent hydrophilic fibers when $We > We_c$. Lateral spreading in hydrophilic fibers is higher for denser and staggered configurations as shown in **Fig.S3a,c**, which explains why these configurations have lower penetration for a given Weber number.

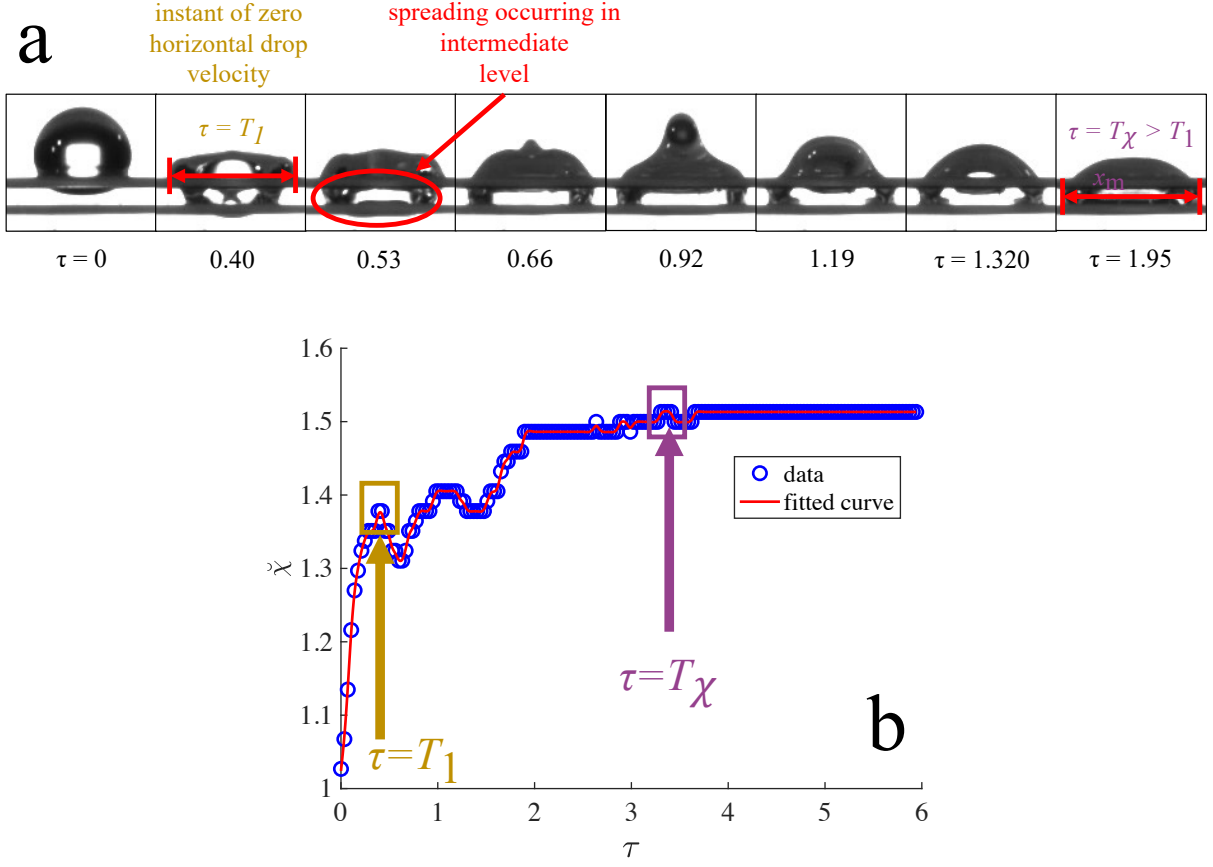


Figure S5: Obtaining the fraction of drop mass infiltration at the first occurrence of local maximum spread using image analysis. A radially symmetric spreading is assumed for the drop section above the fiber array so that disk integration may be used to compute the volume of the drop above the fiber array. The computed volume is then subtracted from the drop's total volume which is computed using D_0 . The result is the infiltration volume which can readily be converted to mass using ρ . In the case where drop spread is asymmetric, the method fails and can lead to a negative value for the mass infiltration, which is excluded from **Fig.S6**. Generally, the ratio of major-to-minor axes for the spherical cap is less than 1.25 ($N = 21$).

The transition between impacting drop advancement and recession is marked by the drop achieving its maximum lateral spreading distance χ_m at a time $\tau = T_\chi$. Maximum lateral spread is generally witnessed on the top surface of the fiber array but occurs on deeper levels for the highest velocity drops. For many of the impact trials onto hydrophilic fibers, $\chi_m \sim \chi_s$, as shown by the interpretive black line in **Fig.S3c**. The lack of contact line pinning enabled by hydrophobicity obstructs any such linear relation between χ_m and χ_s , as seen in **Fig.S3d**. The range of values for $\check{\chi}_m$ is comparable for both coatings as shown by comparison of **Fig.S3c,d**, which is in agreement with previous studies that state maximum spreading is independent of wettability^{2,3}. Penetration $\check{d}_{p,s}$ is more highly correlated to $\check{d}_{p,m}$ (**Fig.7a,b**) than $\check{\chi}_s$ is to $\check{\chi}_m$ (**Fig.S3c,d**) because the horizontal orientation of fibers resists vertical mobility between discretized fiber levels. However, along the horizontal direction, the drop tends to recede into a circular cross-sectional shape in hydrophobic fibers whereas in hydrophilic fibers, the drop remains deformed by the fiber strands through capillarity.

In studies where the impact substrate is non-porous, $\tau = T_\chi$ corresponds to an instant of zero drop kinetic energy which always occurs at the first local maximum in the wetted area⁴, a time we denote as $\tau = T_1$. In our experiments wherein the medium is porous, T_χ does not always coincide with T_1 , as exemplified in Movie S3. In rare occasions, χ_m occurs at times $\tau > T_1$, an example of which is given in **Fig.S5a**. The impacting drop spreads to different extents across the various fiber depth layers as it penetrates, causing the maximum wetted width to occur at an intermediate penetration location. The extent of wetting in an upper layer is not smoothly translated to subsequent layers. The presence of fibers guides the flow of the penetrating drop, creating a virtual disconnection between the liquid trapped at discrete layers. At $\tau = T_1$, the lateral spread velocity of the bulk drop reaches zero before retracting, as shown in **Fig.S5b**.

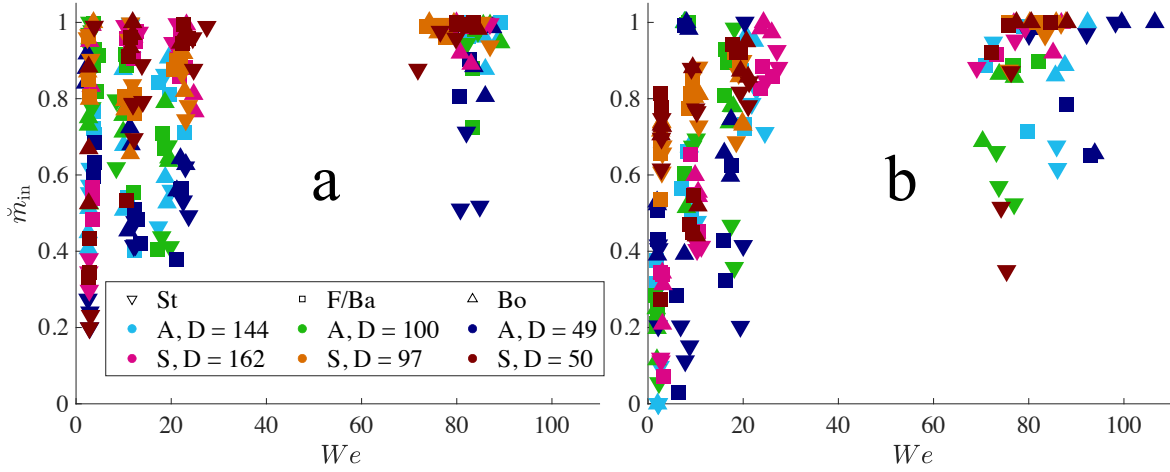


Figure S6: Fraction of mass infiltration at maximum spread in horizontally aligned (a) hydrophilic and (b) hydrophobic fibers as a function of Weber number. Fiber density (D) numbers in strands/cm²; St = Standard orientation, F/Ba = Front/Back orientation, Bo = Bottom orientation; A = Aligned, S = Staggered; refer to **Table 2**.

For a given Weber number, the mass of liquid that penetrates the topmost fiber surface m_{in} at $\tau = T_{max}$ is highly variable and dependent on fiber arrangement. We witness for a fixed Weber number that the ratio $\tilde{m}_{in} = m_{in}/m_0$ varies from 0 to 1, where m_0 is the initial drop mass, as shown for hydrophilic and hydrophobic fibers in **Fig.S6**. As Weber number increases, the variance in $\tilde{m}_{in}$ decreases with $\tilde{m}_{in}$ tending toward unity. The result is that for $We \gtrsim 100$ the entire drop penetrates the fiber array at $\tau \leq T_{max}$. At very low impact velocities ($We < 10$), dense, hydrophobic fibers prevent any measurable penetration, as shown in **Fig.S6b** and as discussed in **Hydrophobicity promotes penetration through fragmentation**. Otherwise, $\tilde{m}_{in}$ values between hydrophilic and hydrophobic fibers are not vastly different, a non-intuitive result based on the results presented in **Hydrophobicity promotes penetration through fragmentation** that show penetration depth is enhanced by hydrophobicity. Using an independent samples *t*-test with a 99% confidence level, the difference in the means of $\tilde{m}_{in}$ between hydrophilic and hydrophobic samples is not statistically significant with *p*-values 0.3626 ($We \approx 20$) and 0.0296 ($We \approx 85$) when $We \gtrsim 15$; otherwise, the hydrophobic mean is lower than its hydrophilic counterpart by 19-41% due to the capillary action with *p*-values 0.0000 ($We \approx 3$) and 0.0010 ($We \approx 10$). The lower the *We*, the greater the difference in the fraction of mass infiltration between hydrophilic and hydrophobic fibers. The expression of greater penetration depth while maintaining a comparable

mass infiltration to their hydrophilic counterparts highlights the promotion of drop fragmentation by hydrophobic fibers and the lateral spreading of drops onto hydrophilic fibers.

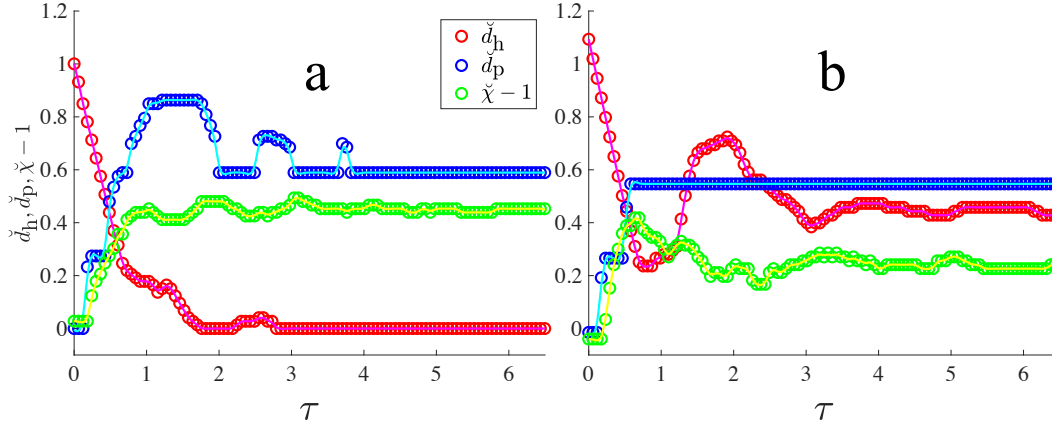


Figure S7: Volume conservation. Normalized dome height, spreading, and penetration depth across inertial time in (a) hydrophilic and (b) hydrophobic fibers corresponding to the image sequences in **Fig.S4**. $We \approx 10.4$ Solid curves are for visualization purposes only.

Volume conservation links dome height, penetration depth, and spreading

An analysis of the drop dome above the fiber array provides insights into the magnitude of drop spreading and penetration. We plot the temporal evolution of normalized dome height $\check{d}_h$, penetration depth $\check{d}_p$, and reduced spreading $\check{\chi} - 1$ for a single trial at $We \approx 10$ onto both hydrophilic and hydrophobic fiber arrays in **Fig.S7a,b**. Drops infiltrating fibers show a temporal dome height $\check{d}_h$ decrease with an increase in penetration depth $\check{d}_p$ and spreading width $\check{\chi}$, a mandate of volume conservation. At low drop impact velocities onto hydrophobic fibers as is the case in **Fig.S7b**, $\check{d}_h$ and $\check{\chi}$ oscillate heavily and erratically, and achieve a comparatively lower $\check{\chi}$ and higher $\check{d}_h$. Movie S4 shows a high-speed video of drops infiltrating hydrophilic and hydrophobic fibers whose descriptive quantities are shown in **Fig.S7a,b**. In hydrophilic fibers, $\check{d}_p$ and $\check{\chi}$ are out of phase whereas $\check{d}_h$ reaches steady-state abruptly and with little oscillation due to the lateral stretching of the drop and contact line pinning. For the same We and D_0 , $\check{d}_{p,s}$ (as opposed to $\check{d}_{h,s}$ and $\check{\chi}_s$) in hydrophobic fibers is generally reached over a shorter inertial time τ than in hydrophilic fibers as shown in **Fig.S8**. The lack of adhesion in the fibers and consequently lack of stability causes the drop to fragment and penetrate quickly to a final depth, preventing vertical oscillation in between fiber strand layers. Additionally, the inability of hydrophobic fibers to hold a spreading drop near its maximum lateral extent results in lateral oscillations and a longer time to reach a motionless drop.

The splitting and capture of a drop body as it impacts a fiber is affected by drop impact velocity⁵⁻⁷, the ratio of the impacting drop to the diameter of the strands⁶⁻¹⁴, and the shape of the strands⁵. We define the achievement of steady-state of any measurable quantity when the change relative to the previously sampled value is within 1% of the previous value and a condition that remains true for all future sampled values. As shown in **Fig.S8**, steady-state penetration, spread, and dome height are achieved more slowly in wedge-shaped fibers in Bottom orientation than in flat-shaped fibers in Standard orientation as the wedge allows the drop to slide and split more

readily in between strands.

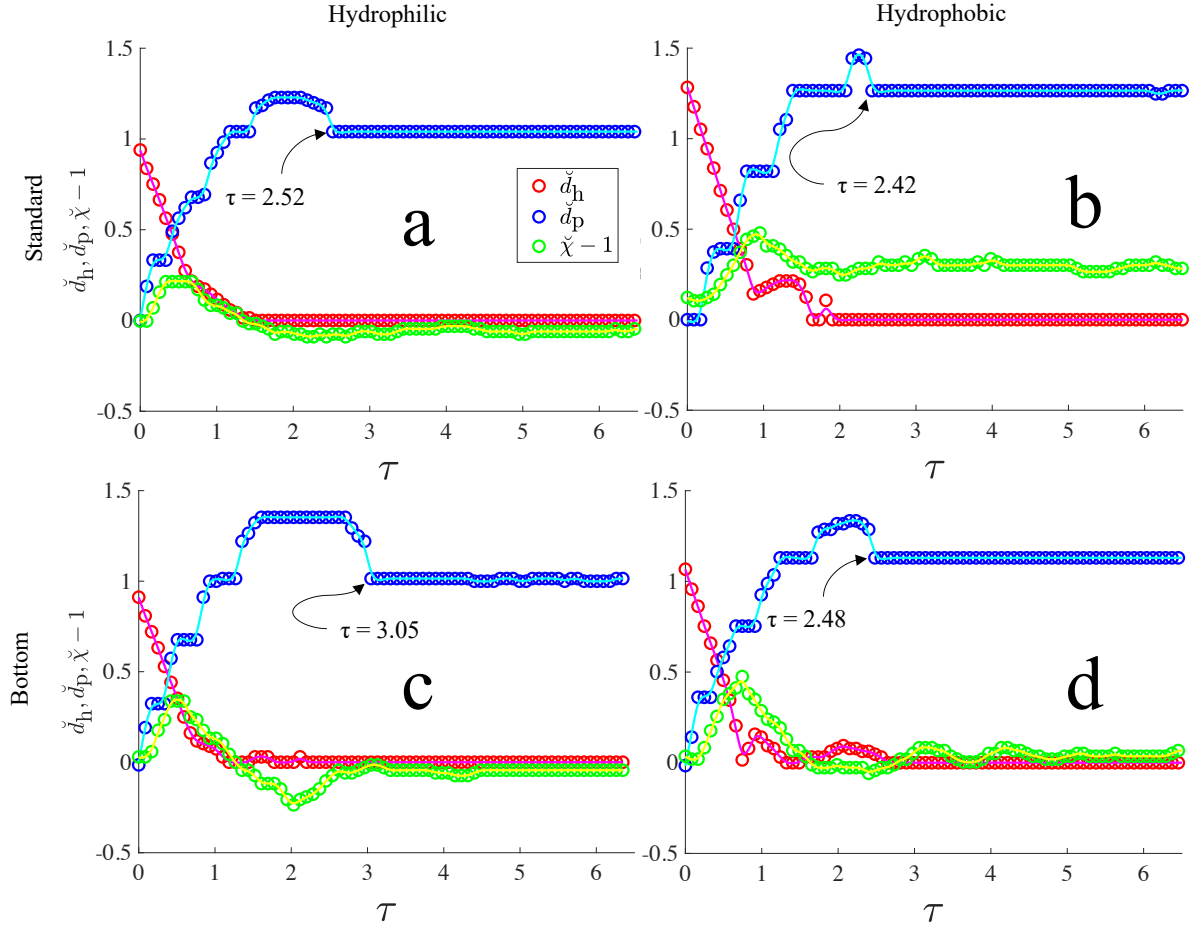


Figure S8: Effect of fiber wettability and strand shape on the evolution of normalized dome height, spreading and penetration depth across inertial time in fibers in (a,b) Standard orientation and (c,d) Bottom orientation; refer to **Table 2**. $We \approx 18.3$ All subplots in the left column use hydrophilic fibers whereas all in the right use hydrophobic fibers. Solid curves are for visualization purposes only.

A decrease in dome height followed by an increase in spreading and penetration is generally observed as a drop impact event manifests through time. However, when considering the steady-state values of different drop impacts, a decrease in steady-state dome height $\check{d}_{h,s}$ does not necessarily lead to an increase in steady-state spreading $\check{\chi}_s$ and penetration $\check{d}_{p,s}$. For instance, $\check{\chi}_s$ is not correlated with $\check{d}_{h,s}$ as shown in **Fig.S9a,b**, whereas $\check{d}_{p,s}$ is negatively correlated with $\check{d}_{h,s}$ for hydrophilic and hydrophobic fibers in **Fig.S9c-d**. A model for the negative relationship between dome height and penetration depth is an area to be explored in future work. Hydrophobic fibers allow the contact line of the droplet to recede as discussed in **Lateral spreading restricts penetration depth, not mass infiltration**. Not only does contact line recession in hydrophobic fibers inhibit spreading as mentioned previously; such recession also tends to increase the variations in the dome height at different impact velocities as can be observed in **Fig.S9b**. Moreover, because the drop is encouraged to maintain its circular shape in hydrophobic fibers, $\check{d}_{h,s}$ values are higher. Additionally, more penetration is needed for $\check{d}_{h,s}$ values in hydrophobic fibers to approach zero because of

reduced spreading.

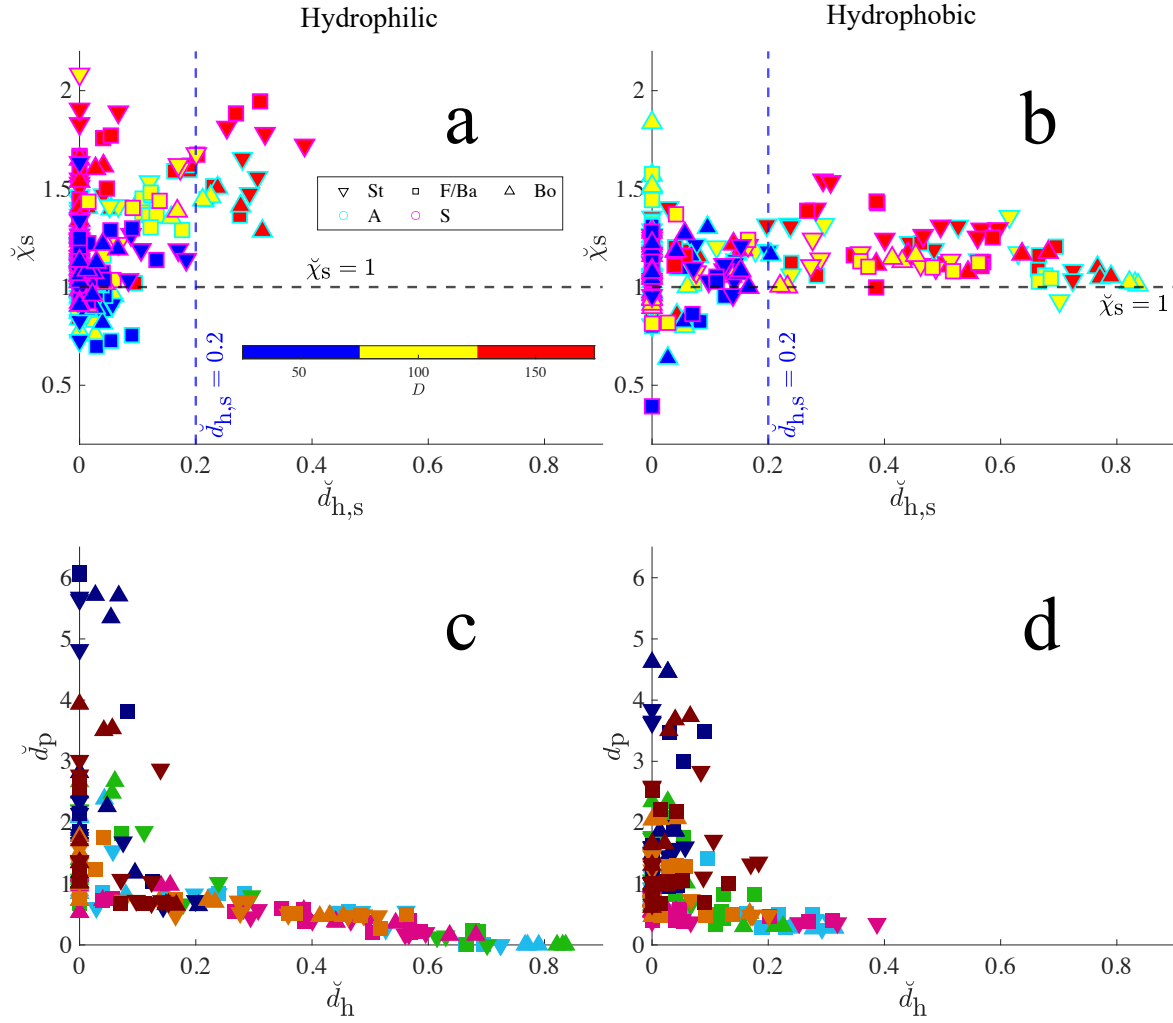


Figure S9: (a,b) Normalized spread versus normalized dome height at steady-state. Blue line roughly demarcates the region to its left where drop impacts on sparse fibers, thus the region where drop fragmentation is likely to occur. (c,d) Plot of normalized penetration depth versus normalized dome height at steady-state. All subplots in the left column use hydrophilic fibers whereas all in the right use hydrophobic fibers. Fiber density (D) numbers in strands/cm²; St = Standard, F/Ba = Front/Back, Bo = Bottom; A = Aligned, S = Staggered; refer to **Table 2**.

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